> **DEFINITION** web3.js is a JavaScript library--commonly referred to as web3--that enables applications to access the services offered by the Ethereum blockchain client node.

This chapter demonstrates the role of web3 and strengthens your knowledge of and skills in application development with web3. You'll take steps toward becoming an informed user of web3 API in developing Dapps. This chapter lets you explore the web3 API in-depth, learning web3 concepts and understanding the essential role that it plays in delivering the blockchain services to Dapps. You'll also learn about various functional modules defined in the API. You'll use web3 to develop a versatile channel feature to build a Dapp for cleaning up global recyclable plastics. In this chapter, you'll see how to use a new concept called side channel for a novel application of micropayments. You'll develop an end-to-end solution for a simplified version of a micropayment channel supported by the Ethereum blockchain. Take time to understand web3 concepts, and carefully follow along with the Dapp development described in this chapter.

> **NOTE** I'll use web3 in this text to refer to the entire library (web3.js). Note that web3 is the class name and web3 is the package name in the web3 JS API codebase in GitHub.

7.1 Web3 API

The web3 API provides a standard set of classes with their functions so that all the participants in a decentralized application can use the same syntax and semantics to interact with the blockchain. Otherwise, the interactions may result in inconsistencies among the participants, rendering the blockchain useless. It may be necessary, for example, to ensure that all participants in an application use the same hash function to